

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 12: Website Traffic Analysis

Aim

To analyze website traffic data using **line chart, bar chart, table, and Tableau dashboard** for understanding page-view trends and click-through rates.

Dataset

Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

Q1. Line Chart — Daily Page Views

Question

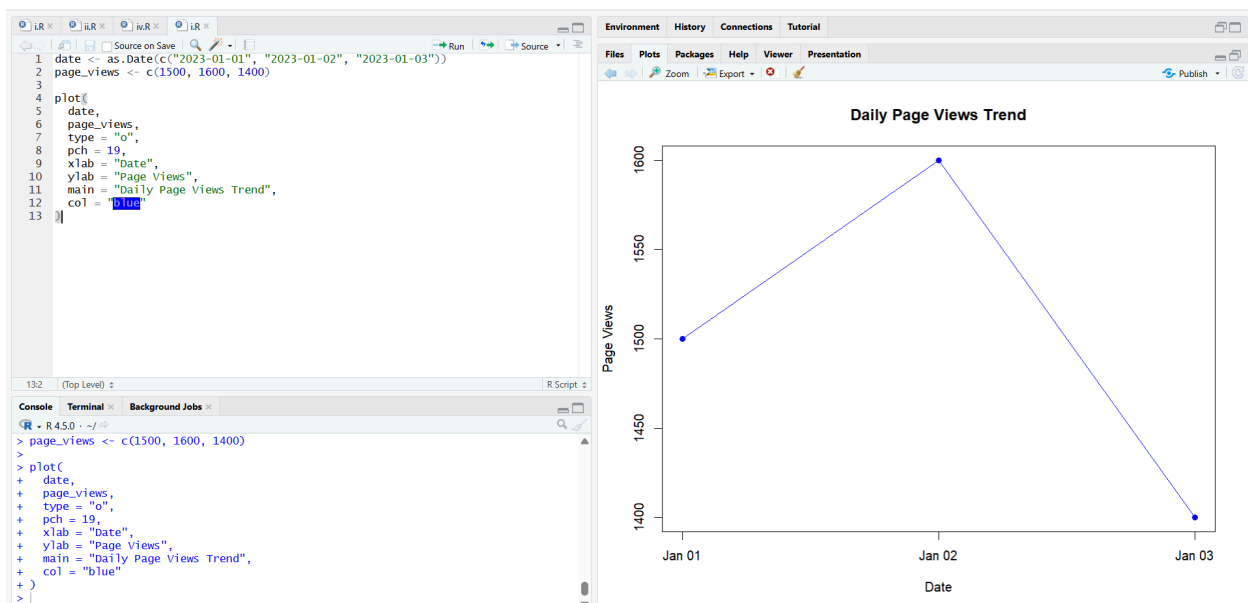
Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R Program

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03"))
page_views <- c(1500, 1600, 1400)

plot(
  date,
  page_views,
  type = "o",
  pch = 19,
  xlab = "Date",
  ylab = "Page Views",
  main = "Daily Page Views Trend",
  col = "blue"
)
```

Output



Q2. Bar Chart — Top N Days by Click-through Rate

Question

Generate a bar chart showing the top N days with the highest click-through rates.

Here, $N = 2$ is used because the dataset contains only three days.

R Program

```
date <- c("2023-01-01", "2023-01-02", "2023-01-03")
click_rate <- c(2.3, 2.7, 2.0)

N <- 2

top_index <- order(click_rate, decreasing = TRUE)[1:N]

top_dates <- date[top_index]
top_rates <- click_rate[top_index]

barplot(
  top_rates,
  names.arg = top_dates,
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  main = "Top 2 Days by Click-through Rate",
  col = "skyblue"
)
```

Output



Observation

January 2 has the **highest click-through rate of 2.7%**, followed by January 1 with **2.3%**.

Q3. Website Traffic Table

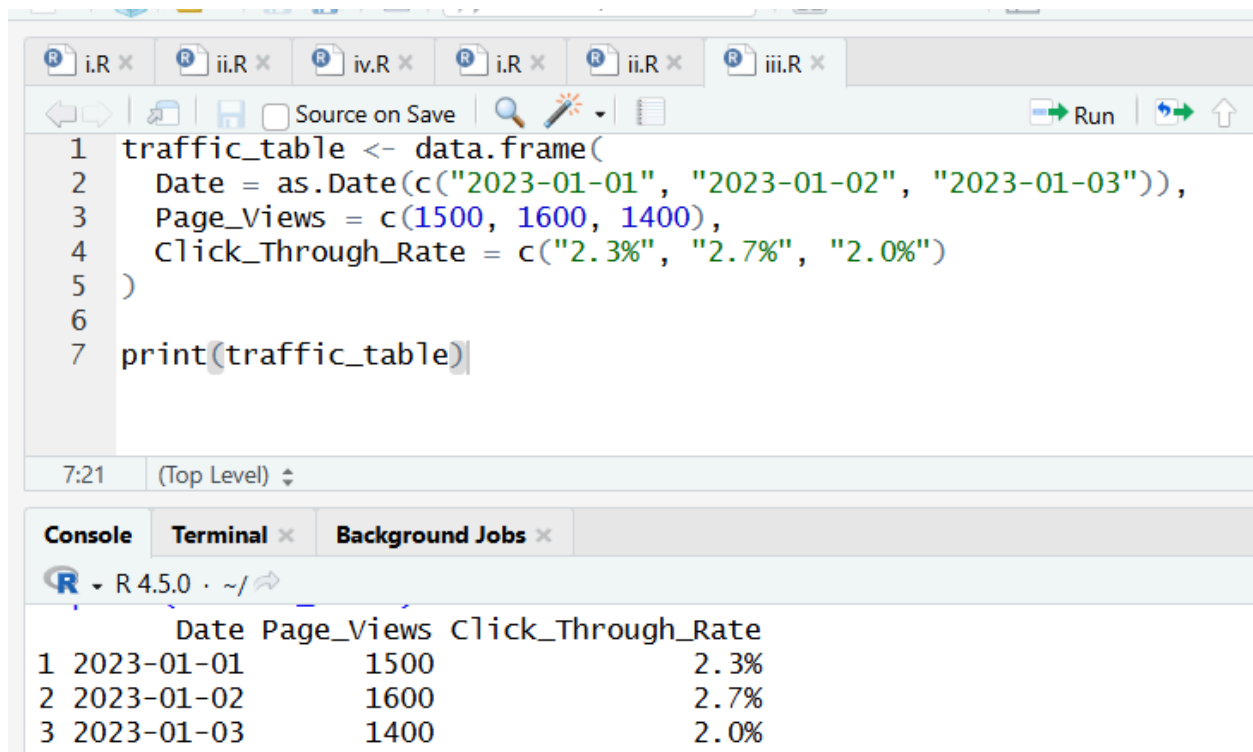
Question

Build a table to show the website traffic data for each day. Label the table elements.

R Program

```
traffic_table <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  Page_Views = c(1500, 1600, 1400),  
  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
)  
print(traffic_table)
```

Output



The screenshot shows the R Studio environment. The script editor contains the following code:

```
1 traffic_table <- data.frame(  
2   Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
3   Page_Views = c(1500, 1600, 1400),  
4   Click_Through_Rate = c("2.3%", "2.7%", "2.0%")  
5 )  
6  
7 print(traffic_table)
```

The console output shows the resulting data frame:

	Date	Page_Views	Click_Through_Rate
1	2023-01-01	1500	2.3%
2	2023-01-02	1600	2.7%
3	2023-01-03	1400	2.0%

Observation

The table successfully displays the complete website traffic information for each day.

Q4. Tableau Dashboard

Question

Develop a Tableau dashboard combining the line chart, bar chart, and table for interactive exploration of website traffic data.

Tableau Procedure

Step 1 — Import Dataset

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**.
3. Import the Website Traffic dataset.
4. Make sure Date is recognized as a **Date** field.

Step 2 — Create Line Chart

- Drag **Date** → Columns.
- Drag **Page Views** → Rows.
- Select **Line** from the Marks card.
- Add the title:

Daily Page Views Trend

Step 3 — Create Bar Chart

- Drag **Date** → Columns.
- Drag **Click-through Rate** → Rows.
- Select **Bar**.
- Sort the bars in descending order.
- Filter to the **Top N** days.

Title:

Top Days by Click-through Rate

Step 4 — Create Traffic Table

Add:

- Date
- Page Views

- Click-through Rate

to the worksheet.

Title:

Website Traffic Data

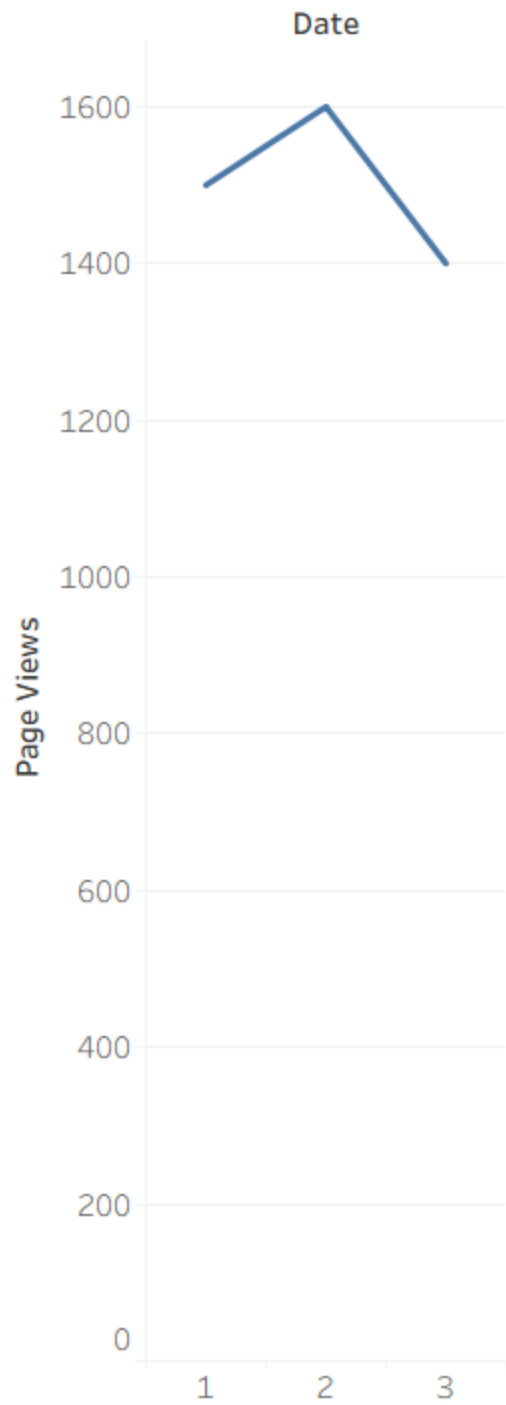
Step 5 — Create Dashboard

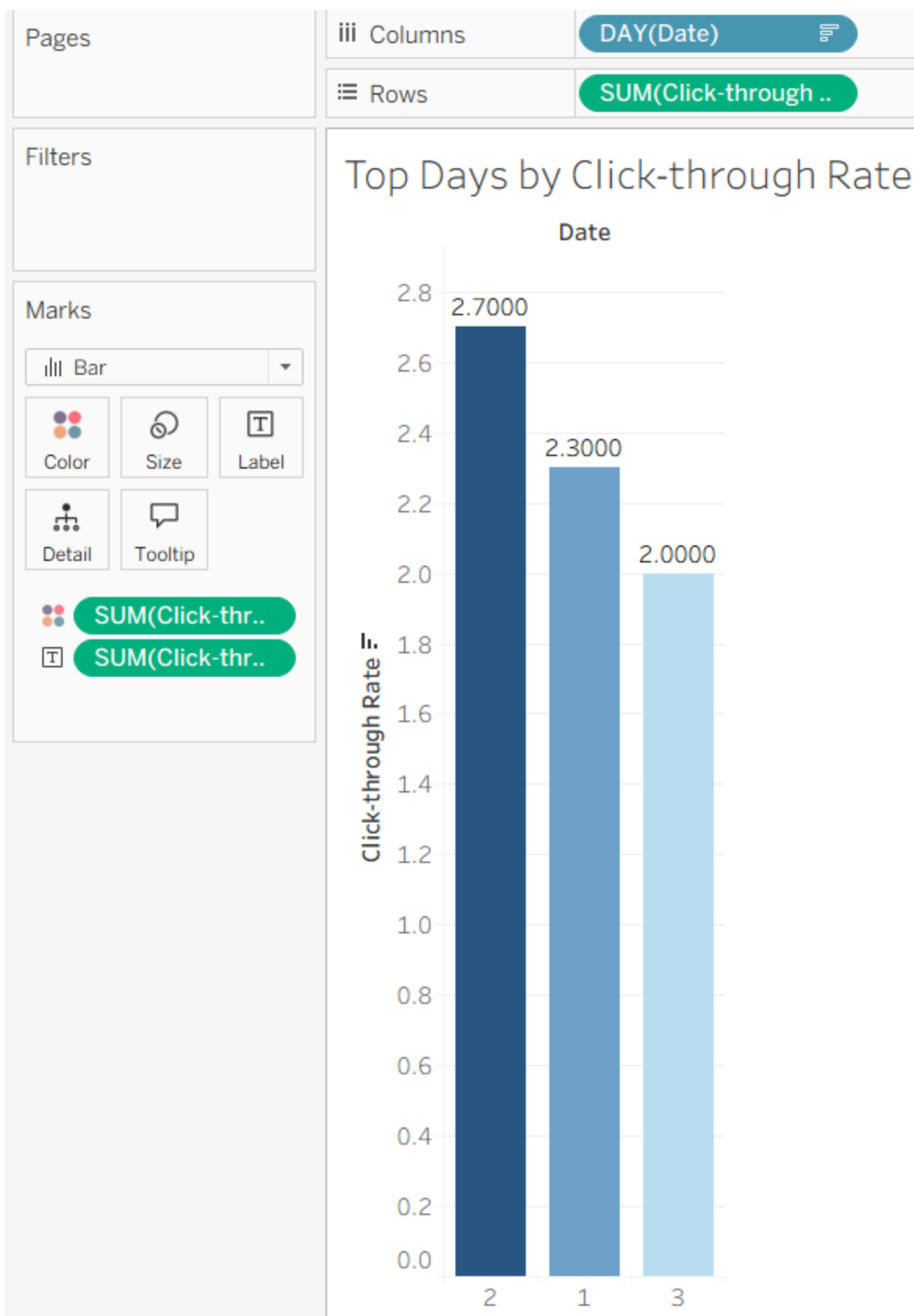
1. Select **New Dashboard**.
2. Add the **Daily Page Views Line Chart**.
3. Add the **Top N CTR Bar Chart**.
4. Add the **Website Traffic Table**.
5. Add suitable filters if required.
6. Arrange and format the dashboard.

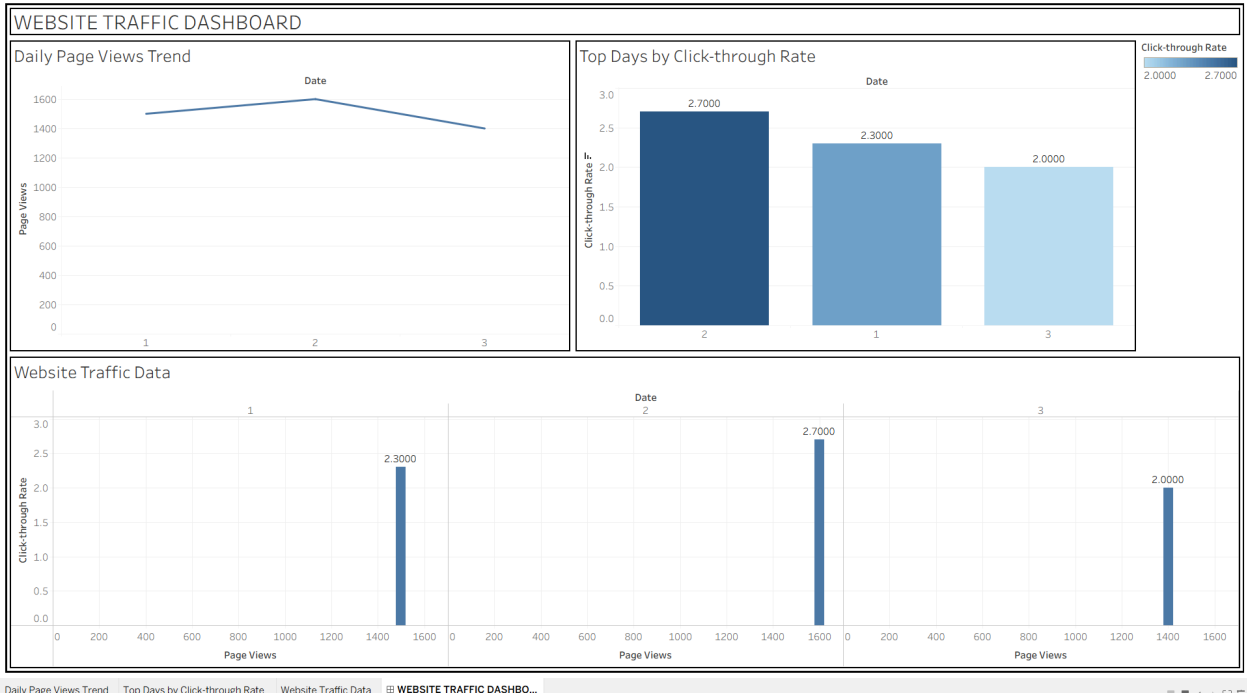
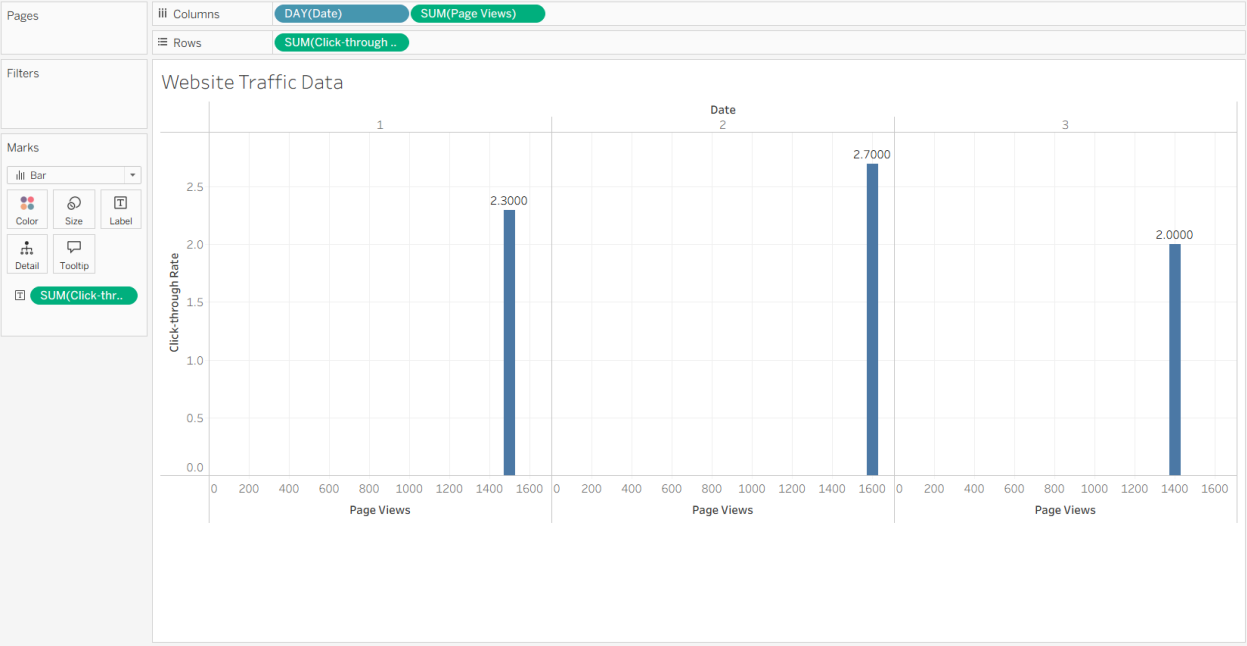
Expected Dashboard

Columns	DAY(Date)
Rows	SUM(Page Views)

Daily Page Views Trend







Procedure

1. Open **RStudio** and enter the given website traffic dataset.
2. Convert the Date values into R Date format using `as.Date()`.
3. Use the `plot()` function to create a **line chart of daily page views**.
4. Use `order()` to identify the days with the highest click-through rates.
5. Use `barplot()` to create a **Top N click-through rate bar chart**.
6. Use `data.frame()` to create a table containing Date, Page Views, and Click-through Rate.
7. Open **Tableau** and import the Website Traffic dataset.
8. Create a line chart for daily page views.
9. Create a bar chart showing the top N days by click-through rate.
10. Create a table displaying the website traffic data.
11. Combine all three visualizations into a **Tableau dashboard**.
12. Add suitable titles, labels, filters, and formatting for interactive exploration.

Result

The website traffic data was successfully analyzed using **R and Tableau**. A line chart was created to visualize daily page views, a bar chart was created to identify the top days by click-through rate, and a table was created to display the website traffic data. These visualizations were successfully combined into an **interactive Tableau dashboard** for exploring website traffic performance.